

Natural Language Processing

CSCI 5832—Lecture 8

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Today

Quiz 1 is Thursday

- Readings and practice quiz on Canvas
- In class, on paper, closed book, one page of notes allowed.

Today

Word meanings and how we might capture them in representations that are useful computationally

- Chapter 5: Embeddings/Vector Semantics
 - Dimensions of word meaning
 - Word embeddings

Question

What role did the word meanings play in our discussion of N-Gram language models or text classification with logistic regression?

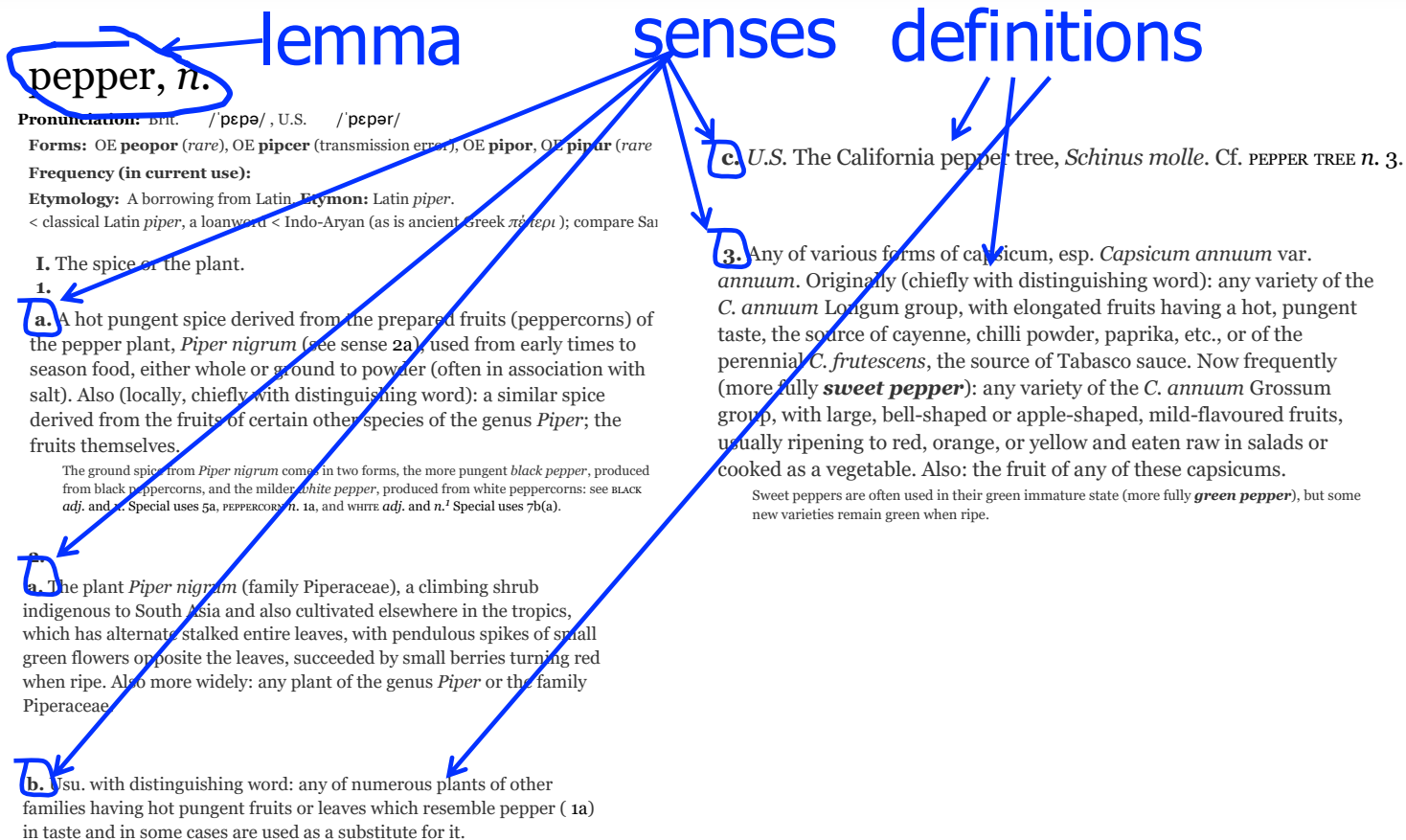
Question

What role did the meaning of the words play in our discussion of text classification with logistic regression?

- Whatever meaning there is is captured via the features we use (like the presence of a word on a list of positive or negative words).
- But the meaning of words like “shower” or “mattress” doesn’t seem to play a role at all.

Seems like we should be able to do more than this.

Words, Lemmas, Senses, Definitions



Senses

Sense refers to a single coherent meaning conventionally associated with a word

- As opposed to its spelling, pronunciation, morphology, lexical category, historical origins, etc.
- Words can have many different senses
 - Some related to each other, some not

Bank (From Wordnet)

Noun

- **S: (n) bank** (sloping land (especially the slope beside a body of water)) *"they pulled the canoe up on the bank"; "he sat on the bank of the river and watched the currents"*
- **S: (n) depository financial institution, bank, banking concern, banking company** (a financial institution that accepts deposits and channels the money into lending activities) *"he cashed a check at the bank"; "that bank holds the mortgage on my home"*
- **S: (n) bank** (a long ridge or pile) *"a huge bank of earth"*
- **S: (n) bank** (an arrangement of similar objects in a row or in tiers) *"he operated a bank of switches"*
- **S: (n) bank** (a supply or stock held in reserve for future use (especially in emergencies))
- **S: (n) bank** (the funds held by a gambling house or the dealer in some gambling games) *"he tried to break the bank at Monte Carlo"*
- **S: (n) bank, cant, camber** (a slope in the turn of a road or track; the outside is higher than the inside in order to reduce the effects of centrifugal force)
- **S: (n) savings bank, coin bank, money box, bank** (a container (usually with a slot in the top) for keeping money at home) *"the coin bank was empty"*
- **S: (n) bank, bank building** (a building in which the business of banking transacted) *"the bank is on the corner of Nassau and Witherspoon"*
- **S: (n) bank** (a flight maneuver; aircraft tips laterally about its longitudinal axis (especially in turning)) *"the plane went into a steep bank"*

Word Relations

One way to get at the meaning of a word is to consider how it relates to senses of other words.

Relation: Synonymy

- Word (senses) that have the same meaning in some or all contexts.
 - filbert / hazelnut
 - couch / sofa
 - big / large
 - automobile / car
 - vomit / throw up
 - Water / H₂O

Relation: Synonymy

There are probably few examples of perfect synonymy.

- Even if many aspects of meaning are nearly identical
 - Frugal vs. tightfisted
- Still may not preserve the acceptability based on notions of politeness, slang, register, genre, etc.
- The Linguistic Principle of Contrast:
 - Difference in form -> Difference in meaning

Relation: Antonymy

Senses that are opposites with respect to one feature of meaning that are otherwise very similar!

dark/light	short/long	fast/slow	rise/fall
hot/cold	up/down	in/out	

- More formally antonyms can either:
 - Define a binary opposition
 - long/short, fast/slow
 - Be *reversives*:
 - rise/fall, arrive/depart, enter/exit

Relation: Word Relatedness

Also called "word associations". Word senses may be related in many ways

- Same "conceptual frame"
 - Buy, sell, product, price, cost, merchant, goods, etc.
 - Fare, route, schedule, timetable, ticket, arrival, departure, reservation
- Part-whole
 - Wheel, fender, roof, axle, tire, engine
- Functional relationships
 - Car/gasoline
- Taxonomically
 - Canid: Coyote, dog, wolf, fox, jackals

Connotation

- Words can have *affective* meanings: aspects of meaning related to emotional state or attitude.
 - Positive/negative emotional state: happy / sad
 - Positive/negative evaluation: great / terrible
 - Positive/negative frame: thrifty / stingy
- As with other word relations, the meanings can be very close even if the connotations are different

Recap

- Words can have multiple senses. And those senses can be distinct and arbitrary or related
- Word senses can be related to the senses of other words along many different dimensions
 - Synonymy
 - Antonymy
 - Similarity
 - Relatedness
 - Taxonomic
 - Connotation

Distributional Hypothesis

Words are defined by their environments (the words that occur around them)

“If A and B have almost identical environments, we say that they are synonyms.”

Harris (1954)

"a word is characterized by the company it keeps"

Firth (1957)

What does *ongchoi* mean?

Suppose you see these sentences:

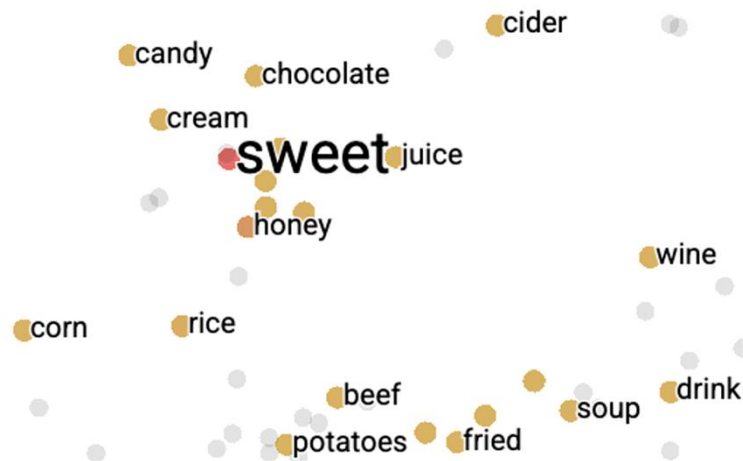
- *Ong choi* is **delicious sautéed with garlic**.
- *Ong choi* is superb **over rice**
- *Ong choi* **leaves** with **salty** sauces
- And you've already seen these:
 - ...spinach **sautéed with garlic over rice**
 - Chard stems and **leaves** are **delicious**
 - Collard greens and other **salty** leafy greens
- Conclusion:
 - Ongchoi is a leafy green like spinach, chard, or collard greens



Meaning Based on Distributional Similarity

Characterize a word's environment, or the company it keeps, as a location in a multi-dimensional space.

- Words nearby are closely related, words far away less so.
- The nature of the relationship is captured by the direction traveled



Word Meanings as Vectors

We can instantiate this idea by capturing a word's meaning as a vector of numbers – where each element represents a location in a multi-dimensional space. This simple representation makes a variety of useful computations possible:

- Similarity measures
- Composition of meaning
- Relational and analogical reasoning
- Visualization and clustering methods

2 Kinds of Embeddings

Sparse

- Original approach from the information retrieval world
- Words are represented as a function of the counts of words they co-occur with, typically at the document level
- Vector dimensionality is often the length of the vocabulary
- Vectors are sparse (lots of 0's)
 - This has advantages and disadvantages

Dense

- Representations trained from larger corpora using semi-supervised learning based on co-occurrence statistics
- Dense: mostly non-zero, floats. Dimensionality varies from hundreds to thousands.
- Typically dimensionality is lower $\ll$ than the vocabulary size.

Term-Document Matrix

Assume we have a collection of documents, contexts, and a vocabulary extracted from that collection. Using this collection, we create a matrix with dimensions $|V| \times |D|$, the size of the vocabulary by the number of documents.

The entries in the matrix are the frequency count of each word in each document.



Term-Document Matrix

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
soldier	1	2	5	25
good	93	76	62	77
wit	21	15	2	3
fool	32	55	2	4

Term-Document Matrix

Both documents and words are represented by vectors of word counts

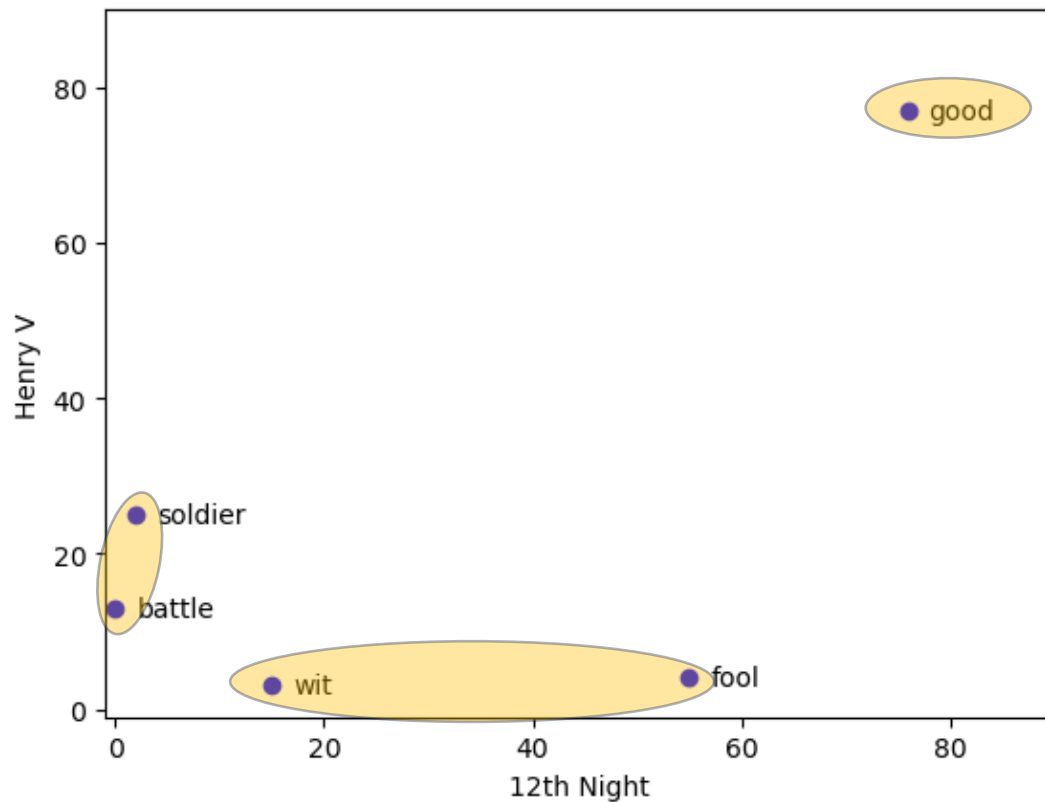
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Term-Document Matrix

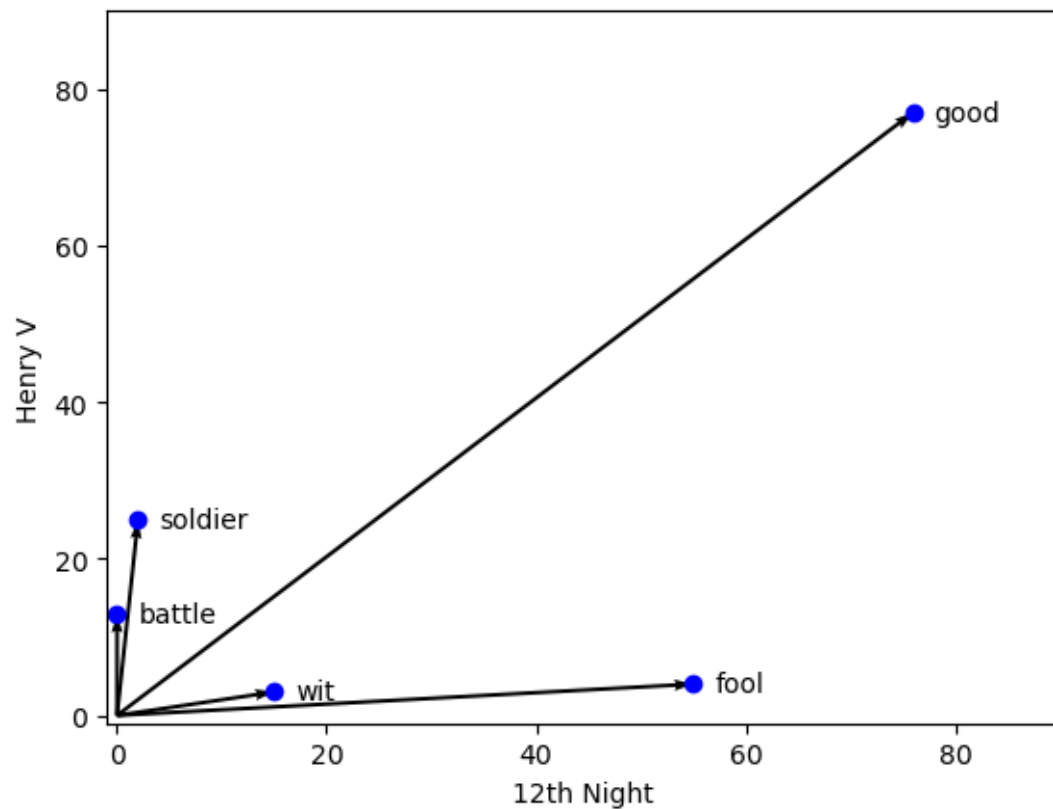
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Visualizing Term Vectors



Visualizing Term Vectors



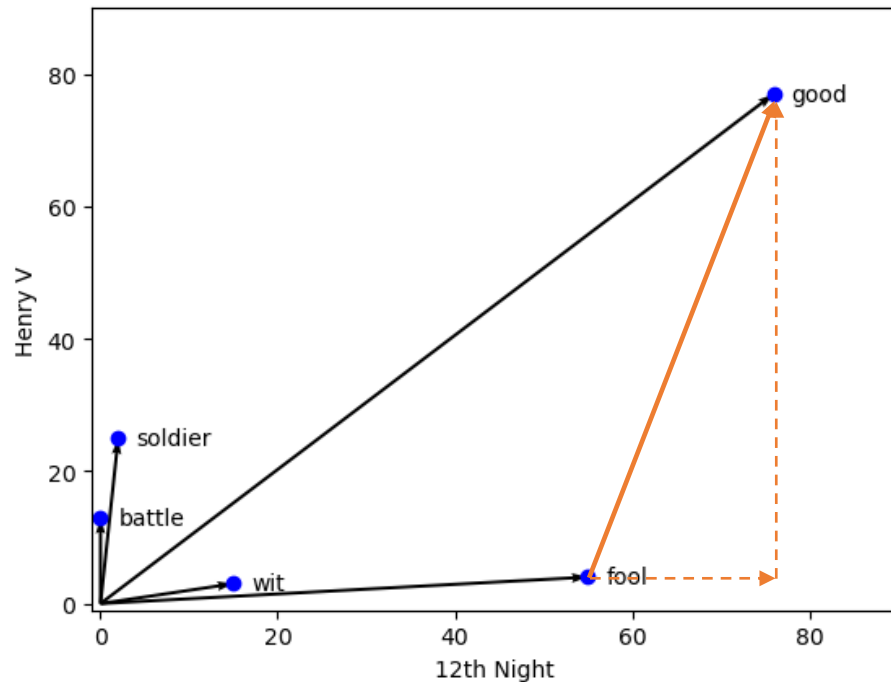
Vector Similarity

Euclidean distance

$$\text{dist}(\mathbf{v}, \mathbf{w}) = \sqrt{\sum_{i=1}^N (v_i - w_i)^2}$$

Dot product

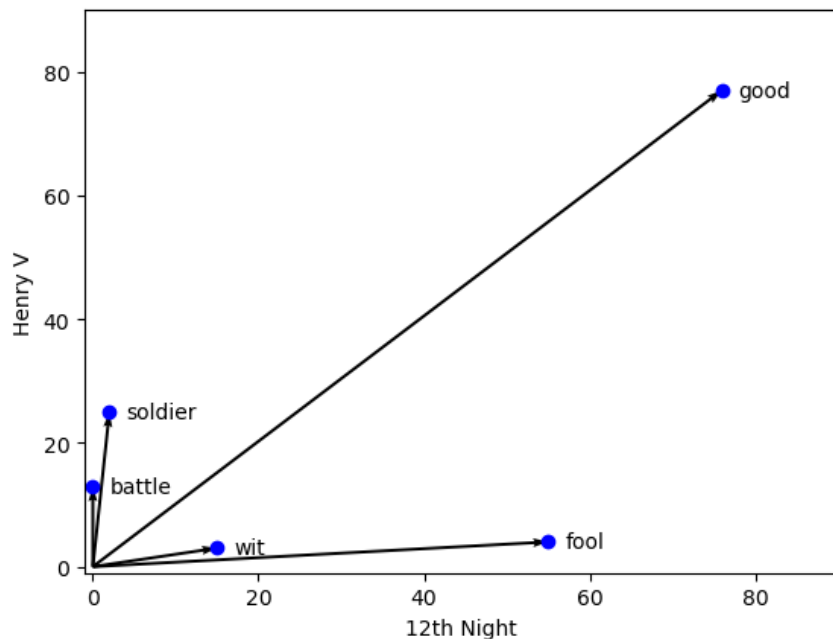
$$\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w} = \sum_{i=1}^N v_i w_i$$



Cosine Similarity

$$\text{CosineSim}(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}| |\mathbf{w}|}$$

$$= \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$



	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
soldier	1	2	5	25
good	93	76	62	77
wit	21	15	2	3
fool	32	55	2	4

How do these words compare based on cosine?

	Battle	Soldier	Good	Wit	Fool
Battle	1	0.95	0.66	0.19	0.10
Soldier		1	0.62	0.20	0.15
Good			1	0.85	0.76
Wit				1	0.91
Fool					1

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
soldier	1	2	5	25
good	93	76	62	77
wit	21	15	2	3
fool	32	55	2	4

How do these documents compare based on cosine?

	AYLI	Twelfth	Caesar	HenryV
As You	1	0.96	0.93	0.89
Twelfth		1	0.82	0.79
Caesar			1	0.97
HenryV				1

Count Transformations

Raw word counts capture useful information about words and their meanings.

- wit and fool frequently appear in the same contexts, as do battle and soldier

However frequent words like good, the, it, or they are not very informative about the context.

- the also frequently appears with fool. But, it also appears frequently with soldier. That doesn't mean that fool and soldier are related.

A better method would compensate for frequent and rare words. We need a transformation of the raw counts that captures this notion. So, instead of raw counts we'll use a function of the counts.

TF-IDF: Term Frequency Inverse Document Frequency



Spark-Jones, 1972

- TF: term frequency $tf_{t,d} = \begin{cases} 1 + \log_{10} \text{count}(t, d) & \text{if } \text{count}(t, d) > 0 \\ 0 & \text{otherwise} \end{cases}$

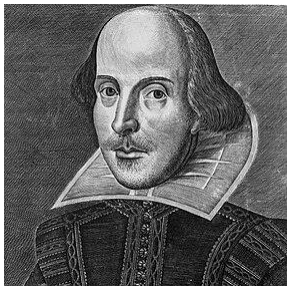
- IDF: Inverse Document frequency
$$\text{idf}_i = \log \left(\frac{N}{\text{df}_i} \right)$$

Annotations for the IDF formula:

 - Blue arrow pointing to N : Total # of docs in collection
 - Blue arrow pointing to df_i : # of docs that have word i

- TF-IDF value for word t in document d :

$$w_{t,d} = tf_{t,d} \times \text{idf}_t$$



$$\begin{aligned}w_{t,d} &= \text{tf}_{t,d} \times \text{idf}_t \\&= (1 + \log_{10}(7)) \times \log_{10}\left(\frac{37}{21}\right) \\&= 0.454\end{aligned}$$

	IDF
battle	0.246
soldier	0.036
good	0
wit	0.037
fool	0.012

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.246	0	0.454	0.52
soldier	0.036	0.047	0.061	0.086
good	0	0	0	0
wit	0.085	0.80	0.048	0.055
fool	0.03	0.03	0.015	0.019

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.246	0	0.454	0.52
soldier	0.036	0.047	0.061	0.086
good	0	0	0	0
wit	0.085	0.80	0.048	0.055
fool	0.03	0.03	0.015	0.019

How do these documents compare based on cosine?

	AYLI	Twelfth	Caesar	HenryV
As You	1	0.37	0.97	0.97
Twelfth		1	0.16	0.17
Caesar			1	0.99
HenryV				1